# GenAI for Engineers — Learner Guide

*IES | v4.0 | Lesson-plan aligned | PPT slides 1–186*

This guide follows the supplied "Lesson Plan with Hands-On Labs (v2 — PUB Feedback)" module sequence, all 16 named activities and learning outcomes. Detailed procedures live here; the presentation uses concise visual explanations. The activity folders are short connected tasks inside the six module time boxes.

## Learning outcomes

| Outcome | Learner evidence |
|---|---|
| LO1 | Explain FMEA, Six Sigma/DMAIC and SPC before applying them. |
| LO2 | Automate engineering documents and analyses with Copilot in Word, Excel and PowerPoint. |
| LO3 | Apply R.I.S.E. and E-MAP with iterative prompting. |
| LO4 | Build and deploy a multi-step Copilot Agent. |
| LO5 | Use Researcher and Analyst for cited research and data analysis. |
| LO6 | Ground an agent on SharePoint with cited, traceable answers. |
| LO7 | Apply the complete 7-Step AI Security Framework. |

## One-day course map

| Time | Module | PPT | Primary activity |
|---|---|---|---|
| 09:00–10:00 | 1 · The Copilot Ecosystem for Engineers | Slides 5–25 | Choose the right Copilot surface; complete access check and AI-vs-engineer challenge. |
| 10:00–11:15 | 2 · Prompting for Engineering Documents | Slides 26–55 | Apply R.I.S.E. and E-MAP, constraints, multi-persona review and iterative prompting. |
| 11:30–13:00 | 3 · Engineering Methodology Primer & Document Automation | Slides 56–94 | Explain and apply FMEA → DMAIC → SPC with three small Copilot in Excel labs. |
| 14:00–15:15 | 4 · Copilot Agents & Microsoft 365 Reasoning Agents | Slides 95–124 | Build and test a multi-step FMEA Agent; use Researcher and Analyst. |
| 15:30–16:15 | 5 · SharePoint Integration with Copilot | Slides 125–143 | Create/inspect a private knowledge base, ground an agent and verify citations. |
| 16:15–16:45 | 6 · AI Security & Engineering AI | Slides 144–156 | Apply all seven security controls |

| | | | |
|---|---|---|---|
| | Toolkit | | and make a release decision. |
| 16:45–17:00 | Closing and Q&A | Slides 157–162 | Toolkit and next action |

## Lesson Plan activity map

| Module | Named activities |
|---|---|
| 1 | Lab 1.1 Access check; Lab 1.2 AI versus Engineer |
| 2 | Lab 2.1 Technical specification; Lab 2.2 Engineering report; Lab 2.3 Risk assessment and method statement |
| 3 | Lab 3.1 FMEA; Lab 3.2 DMAIC with SPC/Cp-Cpk; Lab 3.3 Technical specification pack |
| 4 | Lab 4.1 FMEA Assistant; Lab 4.2 Report Writer; Lab 4.3 Minutes workflow; Lab 4.4 Researcher and Analyst |
| 5 | Lab 5.1 SharePoint knowledge agent; Lab 5.2 Grounded agent chain |
| 6 | Lab 6.1 Seven-step security; Lab 6.2 AI Engineering Toolkit |

# Module 1 — The Copilot Ecosystem for Engineers



Learning outcome: choose the appropriate Microsoft 365 Copilot surface for a required engineering output and state the engineer's verification responsibility.

## AI, GenAI and Microsoft 365 Copilot

Artificial intelligence performs tasks associated with human judgement. Generative AI creates a new draft from instructions and context. Microsoft 365 Copilot provides generative-AI assistance inside Microsoft 365 applications and through Copilot Chat and agents. A fluent answer is still a draft until its source, calculation and meaning are checked.

| Surface | Use it when the output is… | Engineer verifies |
|---|---|---|
| Copilot Chat | A question, brief, plan or comparison | Evidence, assumptions and scope |
| Word | A specification, report, SOP or method statement | Requirements, clauses, numbers and approval |
| Excel | A table, formula, summary or chart | Rows, formulas, denominators and interpretation |
| PowerPoint | A decision story for an audience | Every value and conclusion |
| Agent | A repeatable multi-step output | Instructions, tests, permissions and release |
| SharePoint agent | A cited answer from internal documents | Source passage, revision and access |

## Activity 1.1 — onboarding and access check

- Sign in at https://m365.cloud.microsoft/chat with the approved work or school account.
- Open Word and Excel and confirm the Copilot control is available.
- Open Agents in Microsoft 365 Copilot.

- If a surface is unavailable after two minutes, record the limitation and use the trainer's worked example or paired screen.

## Activity 1.2 — AI versus engineer challenge

Input: "Later pump tests are low. The strainer was replaced. A verification check also occurred. Pilot results improved."

- One group manually separates facts, assumptions and questions.
- One group asks Copilot Chat for the same structure.
- Compare speed, completeness, unsupported claims and unanswered questions.
- Accept only the statements that can be traced to the input.

# Module 2 — Prompting for Engineering Documents



Learning outcome: build and iteratively improve a reviewable engineering prompt using R.I.S.E., E-MAP, constraints, personas and acceptance tests.

## Why prompt engineering improves productivity

Prompt engineering reduces clarification cycles, makes repeat work more consistent, exposes assumptions, and asks for evidence in a format that is faster to review. It cannot compensate for missing or poor-quality source data. The practical goal is a checked result with less rework.

| Principle | Question to answer | Example |
|---|---|---|
| Purpose | What decision or deliverable is needed? | Create a four-row FMEA review. |
| Evidence | Which exact source may be used? | Use only FMEAInput. |
| Context | What system, units and lifecycle apply? | CW-401 delivery test in L/h. |
| Steps | What work must happen and in what order? | Audit → calculate → chart → check. |
| Constraints | What must be preserved or avoided? | Do not alter ratings or claim cause. |
| Acceptance | How will the output be checked? | FM01 RPN must equal 140. |

## R.I.S.E. framework

| Element | Meaning | Engineering use |
|---|---|---|
| R — Role | Perspective and vocabulary | "Act as an FMEA facilitator," not an approving engineer. |
| I — Input | Evidence, rows, revision and scope | Use only the named table and preserve |

| | | |
|---|---|---|
| | | source rows. |
| S — Steps | Ordered calculations and operations | Audit, calculate, compare, present, check. |
| E — Expectations | Format, limits and acceptance test | Show formulas; state unknowns; return a fixed table. |

## E-MAP framework

| Element | Meaning | Example |
|---|---|---|
| E — Engineering | System, function, lifecycle and decision | Chemical dosing skid; standardised delivery test. |
| M — Metrics | Formula, units, target and denominator | Deviation rate = deviations ÷ opportunities. |
| A — Assumptions | Known, unknown and provisional statements | Comparable test conditions are assumed and must be checked. |
| P — Parameters | Limits, revision, operating conditions | Eight-row baseline; TR-401 revision 1. |

# Constraints and acceptance tests

A useful constraint describes observable behaviour. “Be accurate” is difficult to test. “Preserve four supplied rows and ratings, show the RPN formula, flag Severity 9 or higher, and do not approve the design” is specific and reviewable.

| Constraint type | Useful wording |
|---|---|
| Data | Preserve units, time order, missing values and source rows. |
| Evidence | Use only named documents or tables; cite each factual claim. |
| Calculation | Show formulas, source ranges and parameters. |
| Decision | Do not infer root cause, compliance, safety or approval. |
| Output | Return Observation \| Evidence \| Limitation \| Next check. |

## Multiple personas and debate

Personas are useful when each represents a real responsibility. Ask each persona to review the same evidence, challenge another persona, and preserve disagreement rather than forcing consensus.

Run a two-round review. Reliability, Operations and Safety each state one claim with evidence. In round two, each challenges one other claim. Finish with agreed facts, unresolved disagreements and the smallest test that could resolve them. Do not force consensus.

| Persona | Review question |
|---|---|
| Reliability | Which failure mechanism is unsupported? |
| Operations | Which operating or test condition may have changed? |
| Safety | What must be checked before changing equipment or controls? |
| Quality | Which criterion, record or calculation is missing? |

## Iterative prompting procedure

- Run the initial RISE/E-MAP prompt.
- Compare the output with one source passage, row and calculation.
- Select one material error or omission.
- Add one precise instruction that addresses that gap.
- Rerun and compare only the affected section.
- Record the correction and remaining limitation.

## Module 2 document demonstrations

| Output | Copilot surface | Prompt focus | Required check |
|---|---|---|---|
| Technical specification | Word | Requirements, tolerances, tests and clause checklist | Every “shall” traces to an approved requirement |
| Inspection report | Word | Observations, findings, recommendations and executive summary | A finding has a stated criterion |
| Risk assessment + method statement | Word | E-MAP context, hazards, controls and safe sequence | Approved matrix and competent-person sign-off |

| | | | |
|---|---|---|---|
| Management summary | PowerPoint | One headline, chart, evidence, limitation and decision | No value is introduced beyond checked sources |

# Module 3 — FMEA → DMAIC → SPC



Learning outcome: explain each method in plain language, select the correct method for the engineering question, use Copilot in Excel to produce a small output, and independently verify one result.

| Sequence | Question | Primary output |
|---|---|---|
| 1 · FMEA | What could fail and what needs attention? | Prioritised risk review and actions |
| 2 · DMAIC | How do we investigate and improve performance? | Evidence trail, pilot and control plan |
| 3 · SPC | Is process behaviour stable over time? | Control chart and investigation signals |

## FMEA — purpose and use

Failure Mode and Effects Analysis is a structured, proactive review of potential failures. Engineers use it during design, commissioning, process change, operation and periodic review. It helps a multidisciplinary team make risk reasoning visible before a failure causes harm or disruption.

| FMEA field | Question | CW-401 example |
|---|---|---|
| Function | What must the item do? | Deliver liquid |
| Failure mode | How could the function fail? | Low delivery |
| Effect | What happens if it fails? | Incorrect dose |
| Possible cause | Why might it happen? | Restricted strainer |
| Current control | What prevents or detects it? | Delivery test |

# FMEA ratings and RPN

| Rating | Meaning | Review caution |
|---|---|---|
| Severity (S) | Seriousness of the effect | Use the agreed consequence scale. |
| Occurrence (O) | Likelihood/frequency of the cause | Use evidence and a defined scale. |
| Detection (D) | Difficulty detecting before the effect | A higher score means harder to detect. |

RPN = Severity × Occurrence × Detection. For FM01, 7 × 5 × 4 = 140. RPN helps rank discussion. It is not a probability, legal acceptance criterion or safety approval. Apply severity gates separately: FM04 has RPN 90 but Severity 9 and therefore still requires review.

## Acting on FMEA results

- Confirm system boundary, failure logic and rating evidence.
- Rank RPNs and apply severity or regulatory gates.
- Assign an action, owner, due date and required evidence.
- Implement and verify the action.
- Reassess residual S/O/D only after evidence exists.

## Lab 1 — FMEA in Excel (20 minutes)

Open courseware/Labs/Lab-3.1-FMEA-Document/Starter.xlsx. Select a cell in the named table FMEAInput, open Copilot in Excel and paste the prompt from Copilot_Prompt.

```
Act as an FMEA facilitator. Use only FMEAInput. Preserve all rows and supplied ratings. Calculate RPN = Severity × Occurrence × Detection, rank the rows, flag Severity 9 or higher, and add one evidence request and action per row. Show formulas. Do not invent incidents or approve the design.
```

Independent checks: FM01 = 7 × 5 × 4 = 140. FM04 must be flagged because Severity = 9 even though its RPN is lower. Save the ranked table, one evidence request and one proposed action.

# DMAIC — structured process improvement

Six Sigma is a data-based approach for reducing defects and variation. DMAIC is its five-phase roadmap for improving an existing process. It creates a disciplined link from problem definition to sustained control; filling the headings alone does not prove improvement.

| Phase | Question | Output |
|---|---|---|
| Define | What problem matters, to whom and within what boundary? | Problem statement, scope and critical-to-quality measure |
| Measure | How does the process perform now? | Operational definition, data-quality check and baseline |
| Analyse | What patterns and possible causes should be tested? | Prioritised hypotheses and evidence gaps |
| Improve | Which supported change will be piloted? | Controlled change, owner and success criterion |
| Control | How will the gain be monitored and protected? | Metric, frequency, trigger, owner and response |

## Everyday Analyse tools

| Tool | Use | Do not conclude |
|---|---|---|
| Pareto | Order problem categories from largest to smallest to focus effort. | The largest category is the root cause. |
| Fishbone / Ishikawa | Organise possible causes under Machine, Method, Measurement, People, Material and Environment. | Every branch is true. |
| 5-Whys | Develop one causal chain and identify the next evidence check. | The fifth answer is automatically the root cause. |

## Worked rate comparison

Baseline = 36 deviations ÷ 400 opportunities = 9.0%. Pilot = 7 ÷ 200 = 3.5%. The observed change is −5.5 percentage points. Relative reduction is 5.5 ÷ 9.0 = 61.1%. The two measures are different. A short pilot after concurrent changes does not identify cause or prove sustained improvement.

# Lab 2 — DMAIC in Excel (20 minutes)

Open courseware/Labs/Lab-3.2-DMAIC-with-SPC/DMAIC-Starter.xlsx. Select WeeklyData, open Copilot and paste the prompt from Copilot_Prompt.

```
Act as a Six Sigma project assistant. Use only WeeklyData. Calculate Baseline and Pilot deviation rates. Create a five-row Define, Measure, Analyse, Improve, Control summary. Separate facts from hypotheses, show formulas, and state that the short pilot does not prove cause.
```

Independent checks: Baseline 36/400 = 9.0%; Pilot 7/200 = 3.5%. Confirm the output contains exactly five phases, separates observations from hypotheses, and names one next evidence check.

## Pareto worked example

In the optional reference data, Low delivery (55) and Test validity (16) contribute 71 of 93 baseline deviations, about 76%. This helps select a starting point; it does not prove why either category occurs.

## Fishbone and 5-Whys worked example

Possible fishbone branches include Machine: pump wear/valve position; Method: setup/sequence; Measurement: calibration/timing; People: training/handover; Material: fluid condition/contamination; Environment: temperature/supply. A 5-Whys chain may move from low delivery → reduced outlet flow → possible restriction → possible filter loading. Stop and request inspection or differential-pressure evidence before treating the chain as fact.

# SPC — Statistical Process Control

SPC uses measurements in time order to judge whether a process behaves consistently. The control chart distinguishes ordinary common-cause variation from special-cause signals that merit investigation. A signal identifies when to investigate; it does not diagnose the cause.

| Chart element | Meaning |
|---|---|
| Time-ordered points | Measurements plotted in collection order |
| Centre line | Process average for the selected baseline |
| UCL/LCL | Expected process range calculated from baseline variation |
| Signal rule | Agreed pattern that triggers investigation |

## Common cause versus special cause

| Variation | Plain meaning | Engineering response |
|---|---|---|
| Common cause | Small routine fluctuation produced by the current process system | Improve the system if performance is inadequate |
| Special cause | Unusual point or pattern indicating something changed | Check measurement, equipment, conditions and events |

## Control limits versus specification limits

Control limits come from process data and describe behaviour. Specification limits come from design, customer or regulatory requirements and describe acceptability. A 59.5 L/h result can satisfy the fictional 58.9–65.1 L/h specification while still being below the 61.4 L/h lower control limit and unusual for the process.

# Process capability — Cp and Cpk in plain terms

Cp asks whether the estimated stable process spread could fit between the specification limits if centred: Cp = (USL − LSL) ÷ 6σ. Cpk also accounts for where the process mean actually sits: Cpk = min[(USL − mean), (mean − LSL)] ÷ 3σ. A value above 1 means the estimated six-sigma spread fits inside the limits; larger values provide more margin. Interpret only after confirming process stability, measurement quality, distribution and data sufficiency.

## Lab 3 — SPC in Excel (25 minutes)

Open courseware/Labs/Lab-3.2-DMAIC-with-SPC/SPC-Starter.xlsx. Select PumpReadings, open Copilot and paste the prompt from Copilot_Prompt.

```
Act as an engineering data assistant. Use only PumpReadings. From the eight Baseline rows, calculate the mean, sample standard deviation, UCL and LCL. Flag T09–T12, create a line chart with fixed limits, show formulas, and give a three-sentence interpretation. Do not claim a root cause or safety conclusion.
```

Independent check: compare T10 = 59.4 L/h with the calculated LCL. Expected signal rows are T09, T10 and T12 below LCL. The three-sentence interpretation must state the observation, evidence and limitation; it must not identify a root cause or declare the pump safe.

# Module 4 — Copilot Agents and Microsoft 365 Reasoning Agents



Learning outcome: configure and test a reusable multi-step FMEA-to-report workflow, then use Researcher and Analyst with citation and calculation checks.

## When an agent is appropriate

Build an agent when the same bounded task repeats, required inputs and output are stable, acceptance tests exist, and the audience and knowledge permissions are known. Use an ordinary Copilot prompt for one-off or still-ambiguous work.

| Agent component | Course configuration |
|---|---|
| Instructions | Collect → validate → calculate → retrieve → human gate → report |
| Knowledge | FMEA template, rating guide and test cases |
| Stop rules | Ask when S/O/D or source revision is missing; show conflicts |
| Output | Source fields, visible RPN, severity gate, evidence gap, action and audit note |
| Deployment | Preview, publish and share to the approved training audience |

## Guided FMEA Agent build

- Open Microsoft 365 Copilot and select Agents. Choose New agent or open the Copilot Studio agent builder available in the tenant.
- Name the agent “IES Pump FMEA Assistant” and state its narrow purpose.
- Paste the supplied FMEA-AGENT-INSTRUCTIONS.md content.
- Attach the supplied FMEA report template/rating reference if the environment allows knowledge files.

- Test FM01 with S7/O5/D4. Confirm RPN 140, evidence gap and human gate.
- Test a row with Detection missing. The agent must stop and request the value and scoring basis.
- Refine one instruction, rerun both tests, then publish/share only to the approved audience.

# Reasoning agents — Researcher and Analyst

| Agent | Use | Independent check |
|---|---|---|
| Researcher | Multi-source standards/technology comparison with citations | Open one citation and compare the passage with the claim |
| Analyst | File-based calculations, charts and data interpretation | Reproduce important formulas and denominators |

## Researcher prompt

Act as an engineering research assistant. Compare the supplied recording requirements with current approved guidance. Separate supplied facts from external findings. Cite each claim and list unresolved differences for engineering review.

## Analyst prompt

Use only the six rows in WeeklyData. Check row count, phases and missing values. Calculate Baseline and Pilot rates, percentage-point change and relative reduction. Create a comparison chart, show formulas, state one limitation and one next evidence check. Do not claim cause or sustained improvement.

Independent checks: 9.0%; 3.5%; −5.5 percentage points; 61.1% relative reduction.

# Module 5 — SharePoint Integration with Copilot



Learning outcome: organise a private engineering knowledge folder, ground an agent on that folder, test supported and missing-evidence questions, and pass only checked excerpts to another agent.

## Prepared environment

| Item | Value |
|---|---|
| Site | https://tertiaryinfotech.sharepoint.com/sites/IES-GenAI-for-Engineers |
| Library / folder | Documents / Engineering Knowledge Base |
| Sources | TR-401-Training-Extract.pdf; FMEA-Starter.xlsx; DMAIC-Starter.xlsx; SPC-Starter.xlsx |
| Agent | IES Engineering Knowledge Assistant |

## Create or inspect the knowledge base

- Confirm the site is private and inspect owners, members and visitors.
- Open Documents / Engineering Knowledge Base.
- Confirm exactly four synthetic sources and their revisions.
- Create or inspect the folder-scoped agent.
- Use instructions that require citations, separate facts from interpretation, and preserve missing evidence.

## Agent test cases

| Test | Expected result |
|---|---|
| Which fields must TR-401 record? | Test ID, timestamp, volume and duration, with citation to clause 1. |

| | |
|---|---|
| What approved delivery limit does TR-401 state? | Not specified; request the approved design requirement. |
| Compare the three workbook purposes. | FMEA risk review, DMAIC improvement summary, SPC monitoring; citation to each file. |

## End-to-end integration

Ask the SharePoint agent for a requirement. Open and verify the cited passage. A named engineer approves the excerpt as context. Pass only that checked excerpt to the FMEA or Report Writer agent. The generated document shows the source and reviewer. This preserves the human gate between retrieval and document generation.

# Module 6 — 7-Step AI Security Framework



Learning outcome: apply every control to the Module 5 grounded answer and make an explicit approve, revise or reject decision.

| Step | Action | Evidence |
|---|---|---|
| 1 · Data classification | Classify input/output using organisational policy | Classification and prohibited fields |
| 2 · Governance & access | Check source, user, agent scope and audience | Site/folder permissions and revision |
| 3 · Secure prompting | Minimise sensitive data; resist embedded instructions | Final prompt and redactions |
| 4 · Output verification | Open citation; reproduce calculation; check units | Pass/correction/unknown |
| 5 · Human sign-off | Name competent reviewer and decision | Reviewer, date and conditions |
| 6 · Traceability & audit | Retain prompt, sources, output and corrections | Reproducible record |
| 7 · Awareness & incident response | Detect, stop, contain, report and improve | Incident path and retest |

## Applied review

- Use the SharePoint-agent workbook comparison and TR-401 missing-limit answer.
- Complete all seven rows in AI-SECURITY-EVIDENCE-LOG.csv.
- Open one citation and reproduce one FMEA, DMAIC or SPC calculation.
- Identify the embedded prompt-injection attempt as untrusted document content.
- Make one release decision: approve for named audience, revise and recheck, or reject and escalate.

# Course close — personal AI Engineering Toolkit

| Keep | Contents |
|---|---|
| Prompt patterns | RISE, E-MAP, constraints, personas and acceptance checks |
| Method quick references | FMEA → DMAIC → SPC |
| Agent assets | Instructions, knowledge and test cases |
| Grounding checks | Folder scope, citations and missing-evidence test |
| Security gate | Seven-step evidence log and release decision |

## One next action

Choose one small repeatable workflow. Name its evidence, required output, acceptance test and approving owner. Trial it with synthetic or approved data before introducing live confidential information.

## Microsoft 365 learning links

- Microsoft 365 Copilot: https://m365.cloud.microsoft/chat
- Copilot in Excel: https://support.microsoft.com/en-us/office/get-started-with-copilot-in-excel
- Copilot in Word: https://support.microsoft.com/en-us/office/welcome-to-copilot-in-word
- Copilot in PowerPoint: https://support.microsoft.com/en-us/office/welcome-to-copilot-in-powerpoint
- Microsoft 365 Copilot agents: https://support.microsoft.com/en-us/topic/agents-in-microsoft-365-copilot
- Researcher: https://learn.microsoft.com/en-us/microsoft-365/copilot/researcher-agent
- Analyst: https://support.microsoft.com/en-US/Microsoft-365-Copilot/get-started-with-analyst-in-microsoft-365-copilot
- SharePoint agents: https://support.microsoft.com/en-us/office/get-started-with-agents-in-sharepoint

## File map

| Resource | Path |
|---|---|
| Slides | courseware/Slides/IES-GenAI-for-Engineers.pptx |
| All 16 Lesson Plan activities | courseware/Labs |
| FMEA workbook | courseware/Labs/Lab-3.1-FMEA-Document |
| DMAIC + SPC workbooks | courseware/Labs/Lab-3.2-DMAIC-with-SPC |
| Agent/SharePoint/security files | courseware/Trainer-Demonstrations |
| Optional quick references | PPT slides 163–186 |